

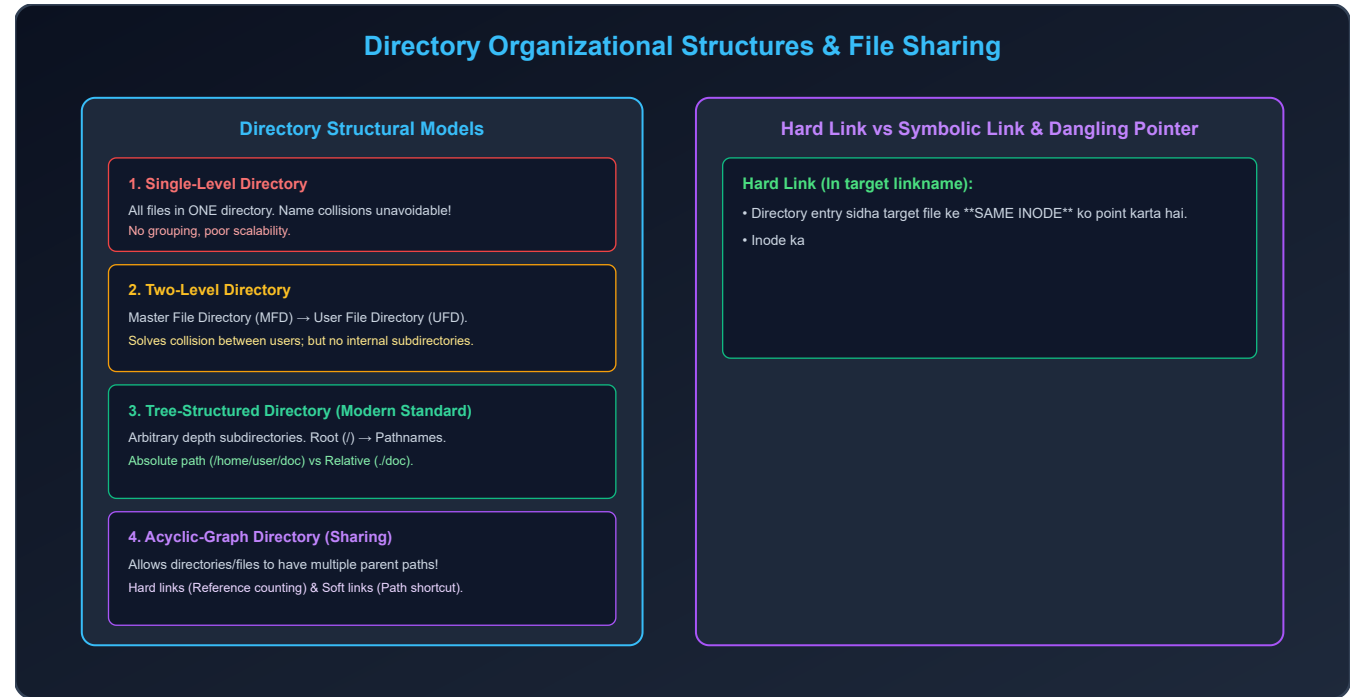
# Module 08: Directory Structures and File Sharing

## 1. Directory Structures Evolution

Single-Level (colliding) → Two-Level (MFD/UFD) → Tree-Structured (hierarchical) → Acyclic Graph (shared files via links).

## 2. Hard Link vs Soft Link (UNIX)

| Feature          | Hard Link                                  | Soft / Symbolic Link                           |
|------------------|--------------------------------------------|------------------------------------------------|
| Inode            | Same Inode number share karta hai          | Naya Inode create hota hai (contains pathname) |
| Target Deletion  | File intact rehti hai (until link_count=0) | <b>Dangling broken pointer</b> ban jata hai    |
| Cross-Filesystem | No                                         | <b>Yes</b>                                     |



link\_count increment ho jata hai (e.g., count = 2). • Original file delete karne par bhi file disk se delete nahi hoti! • File tabhi delete hoti hai jab link\_count == 0 ho jaye! • Limitation: Cannot cross filesystem boundaries; no directory hard links. Soft / Symbolic Link (ln -s target linkname): • Ek nayi file banti hai jiska apna ALAG INODE hota hai. • Is file ke andar target file ka pathname store hota hai. • Filesystems cross kar sakta hai aur directories par link ban sakta hai. • The Dangling Pointer Problem: Agar original file delete ho jaye, to soft link ek non-existent path ko point karta rehta hai ("Broken / Dangling Link")!